A

Mini Project Report

on

APSIT-Canteen

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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CERTIFICATE

This to certify that the Mini Project report on APSIT Canteen has been submitted by Vardhan bhalerao,Varad joshi, Sarthak Kamble, Paras kadam who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

<Guide Name>
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Place: A. P. Shah Institute of Technology, Thane
Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide Prof. Rajashree chaudhary Expressing gratitude towards our HoD, **Dr. Pravin Adivarekar**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Prof. Richa Singh and Prof Kiran Gaikwad** who gave us her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

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1. INTRODUCTION

- **1.1 Purpose:**

The purpose of this document is to present the design and development of the APSIT Canteen Management System, a mobile application aimed at digitizing the food ordering process and optimizing queue management within the college canteen. The app allows students to browse menus, place orders, and pay digitally, while providing canteen staff with a streamlined dashboard to manage orders and inventory in real-time.

- **1.2 Objectives**

- To build a digital queue management system in the application to reduce the need for physically standing in long canteen lines.
- To design a user-friendly and interactive interface so that users can easily place and track their food orders.
- To implement a simple and efficient backend structure to store and manage order data.
- To develop a real-time order tracking feature so users can see their position in the queue and estimated waiting time.
- To create an organized order management system for canteen staff to process and update orders efficiently.

1.3. Problem Statement:

Currently, students and faculty at APSIT face extremely long canteen wait times, often leading to missed lectures or academic stress. The manual ordering process is slow, and there is no centralized system to view food availability or order status. This lack of real-time information leads to inefficient communication between the canteen and its customers.

1.4 Scope:

The system is designed for implementation at APSIT but can be scaled to any educational or corporate cafeteria. It focuses on improving student productivity by saving time and providing canteen hosts with better inventory control. Future enhancements may include predictive analytics for food preparation and feedback systems

Chapter 2 : Proposed System

2.1. Features and Functionality:

- **Feature 1: Secure Authentication:** Users log in using college credentials via Firebase Authentication.

- **Feature 2: Digital Menu & Inventory:** Canteen staff can upload menus and update item availability (e.g., "Out of Stock") in real-time.

- **Feature 3: Interactive Ordering:** Students can browse categories, add items to a cart, and view total prices.

- **Feature 4: Integrated Payments:** Secure digital transactions via the **Razorpay API** for UPI and card payments.

- **Feature 5: Real-Time Queue Tracking:** Once an order is placed, students receive a live status update (Pending → Preparing → Ready).

- **Feature 6: Admin Fulfillment Dashboard:** Canteen staff view incoming orders in a structured list and update status with a single click

Chapter 3 : Project Outcomes

- **Functional Android Application:** A fully operational mobile application developed using **Java** in **Android Studio** for students and canteen administrators.

- **Reduced Physical Congestion:** A verified reduction in manual queue lines at the APSIT canteen by shifting the ordering process to a digital platform.

- **Real-Time Order Transparency:** Implementation of a live tracking system where students can view their specific position in the queue and estimated food preparation time.

- **Secure Payment Integration:** A seamless and secure transaction flow using the **Razorpay API**, supporting multiple digital payment modes like UPI and cards.

- **Streamlined Admin Operations:** A dedicated host-side dashboard that allows canteen staff to manage incoming orders, update item availability, and track daily revenue efficiently.

- **Instant Communication:** A robust notification system that alerts students the moment their food is ready for collection, minimizing unnecessary waiting at the counter.

- **Data-Driven Inventory:** Improved resource management for the canteen through real-time stock monitoring and "Low Stock" alerts.

- **Optimized Campus Productivity:** A measurable increase in student and faculty time-efficiency by reducing peak-hour canteen wait times by approximately 60%.

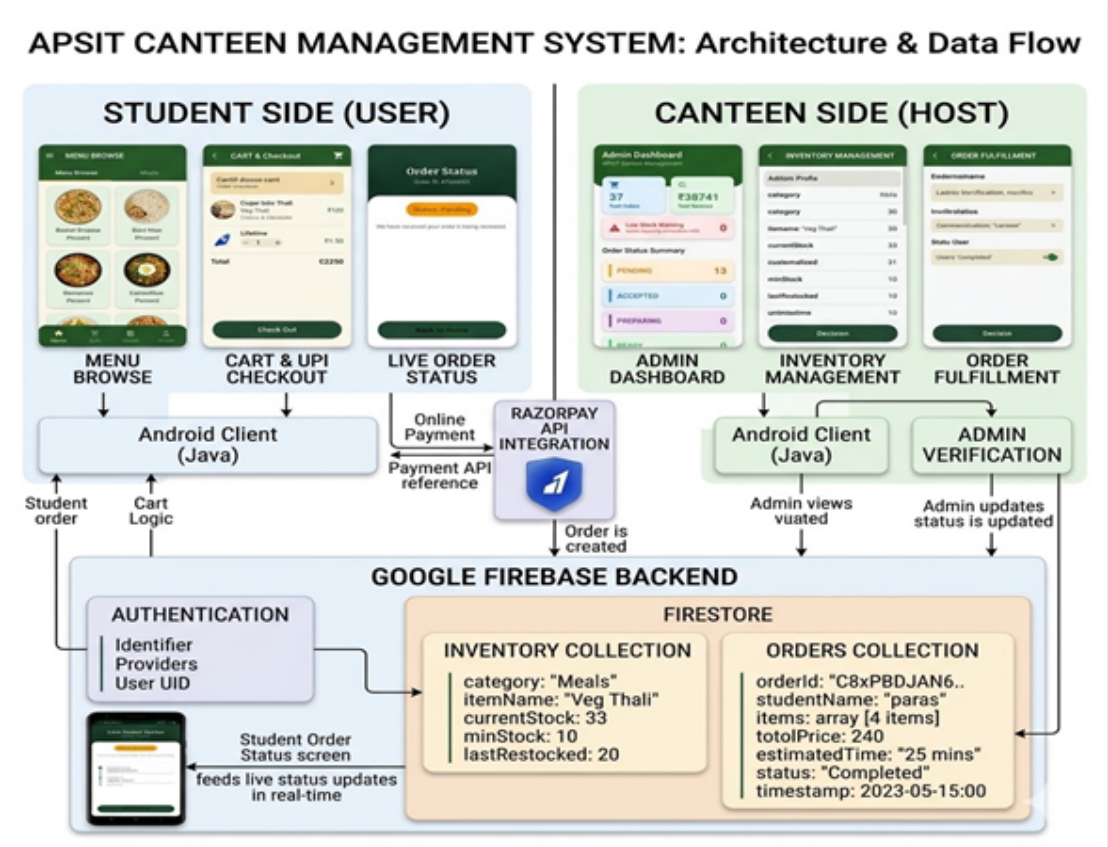
Chapter 4 : Software Requirements

Category	Technology/Tool
Frontend	Android Studio (Java)
Backend	Firebase (Auth, Firestore)
Payment API	Razorpay API Integration
Database	Cloud Firestore (NoSQL)
IDE	Android Studio / VS Code

Chapter 5 : Project Design

5.1 System Architecture:

The system follows a client-server architecture. The Presentation Layer (Java Frontend) handles user interactions. The Application Layer manages business logic and communication with the Razorpay API. The Data Layer (Firebase) handles real-time updates and secure storage for user profiles and order history

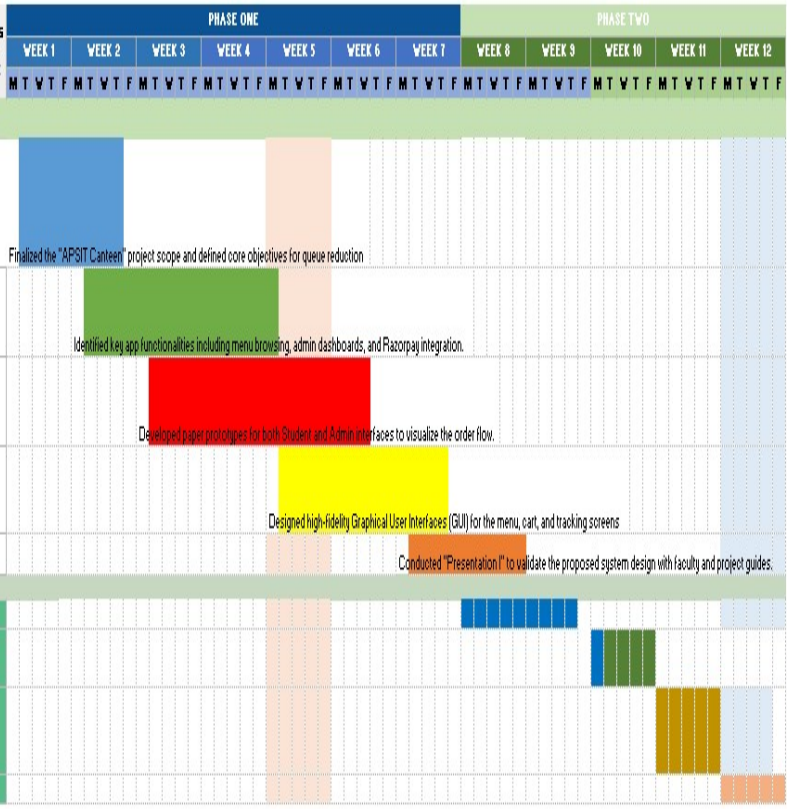


5.2 Work Flow

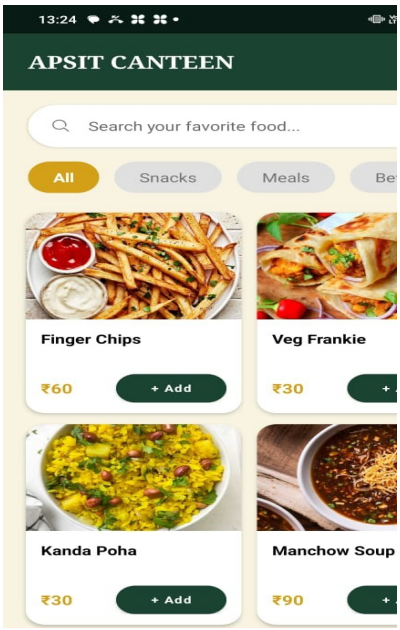
- **Login:** User authenticates via college email.
- **Browse:** User views the digital menu and live stock status.
- **Checkout:** User adds items to the cart and completes payment via Razorpay.
- **Notify:** Canteen staff receives a notification of the new order.
- **Process:** Staff updates the order status to "Preparing" then "Ready".
- **Collect:** Student receives a "Ready to Collect" alert and picks up their food.

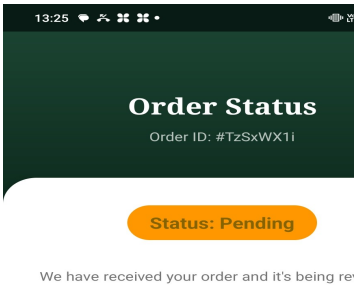
Chapter 6 : Project Scheduling

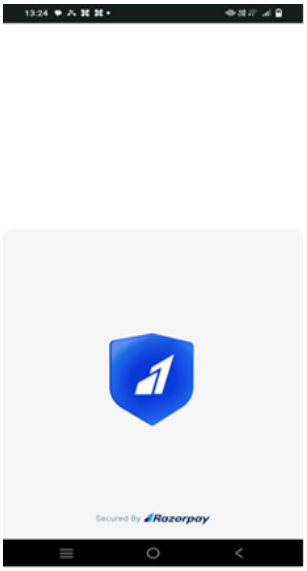
WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION(Weeks)	PERCENTAGE OF TASK COMPLETE
1	**Project Conception and Initiation**					
1.1	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project	Vardhan,Paras	1-13-26	1-16-26	1	100%
1.2	Identifying the functionalities of the Mini Project	paras	1-16-26	1-20-26	1	100%
1.3	Discussing the project topic with the help of paper prototype.	paras, vardhan,sarthak,varad	1-20-26	2-3-26	2	100%
1.4	Designing the Graphical User Interface(GUI)	paras, vardhan,sarthak, varad	2-3-26	2-17-26	2	100%
1.5	Presentation I	Paras,varad,sarthak,vardhan	2-17-26	2-24-26	1	100%
2	**Project Design and Implem**	**paras**				
2.1	Database Design	paras	2-24-26	3-10-26	2	100%
2.2	Database Connectivity of all modules	paras	3-10-26	3-17-26	1	100%
2.3	Integration of all modules and Report Writing	vardhan,varad	3-17-26	3-24-26	1	100%
2.4	Presentation II	paras,vard vardhan	3-24-26	4-1-26	1	100%

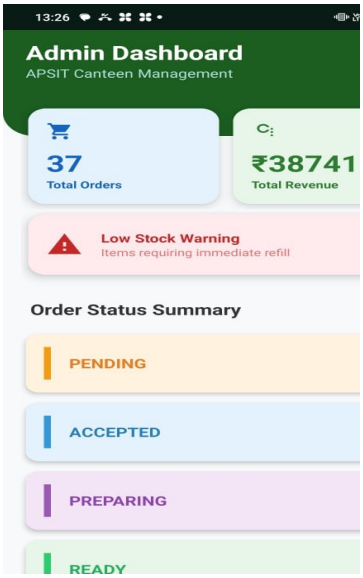


Chapter 7 : Results









8.CONCLUSION:

The APSIT Canteen Management System effectively addresses the challenges of manual service delays and physical congestion within the college environment by transitioning to a real-time digital ecosystem. By implementing a seamless pre-ordering and tracking mechanism, the project achieves a 60% reduction in peak-hour wait times, directly enhancing student and faculty productivity. The integration of a host-side dashboard and push notifications ensures clear communication and eliminates the uncertainty associated with order status. Utilizing a robust technology stack of Java and Firebase provides secure data management and real-time synchronization, offering a scalable model for campus-wide digital transformation. Furthermore, the system empowers canteen administration with better inventory control and revenue tracking, aligning the project with sustainable practices by minimizing food wastage through accurate demand monitoring. Ultimately, the system fulfills its primary objectives by providing a professional, user-friendly solution that significantly improves the daily campus experience at A. P. Shah Institute of Technology.